

Evaluation of the Performance of Neural Network-Based Surrogate Models Applied to Multi-Objective Optimization Problems in Bioinformatics

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Abstract. Flux Balance Analysis (FBA) is widely used for metabolic engineering optimization but becomes computationally expensive when coupled with multi-objective evolutionary algorithms. This work evaluates Artificial Neural Network (ANN) surrogate models for predicting biomass growth and tryptophan production in *Escherichia coli* using the iML1515 metabolic model. Two input representations were investigated: integer encoding with embeddings and multi-hot encoding. To address metabolic discontinuities caused by lethal gene knock-outs, hybrid classification-regression pipelines combining Support Vector Machines and ANNs were developed. Results show that the proposed cross-representation hybrid architecture achieved the best overall performance, reaching global R^2 values of 0.7547 for biomass and 0.7414 for tryptophan production. The study demonstrates that separating viability prediction from flux regression improves surrogate modelling of genome-scale metabolic networks and provides a promising framework for future surrogate-assisted evolutionary optimization.

Keywords: Surrogate Models, Flux Balance Analysis, Machine Learning, Multi-Objective Optimization, Metabolic Engineering.

1 Introduction

In many real-world applications, particularly in bioinformatics, the evaluation of candidate solutions relies on computationally expensive simulation models, making the optimization process increasingly costly when combined with Multi-Objective Evolutionary Algorithms (MOEAs), since these algorithms require a large number of function evaluations across successive generations (Song et al., 2025).

In this work, the optimization of tryptophan production in *Escherichia coli* is selected as a case study. This problem involves multiple conflicting objectives, maximizing cell growth and amino acid production while minimizing resource consumption, and relies on Flux Balance Analysis (FBA) for solution evaluation, a simulation method that requires solving a linear programming problem at each call. Given the high number of

evaluations demanded by evolutionary optimization, the repeated use of FBA becomes computationally prohibitive, making this an ideal and practically relevant scenario for the application of surrogate-assisted multi-objective optimization.

Multi-objective optimization problems involve the simultaneous optimization of conflicting objectives, leading to a set of trade-off solutions known as the Pareto front (Coello et al., 2007; Deb, 2001; Hu et al., 2017). This computational burden represents a major bottleneck in practice, and motivates the use of surrogate models as a more efficient alternative.

Surrogate models, also known as meta-models, are trained to approximate the original evaluation function, providing faster predictions while maintaining acceptable accuracy (Wang et al., 2026). In the context of multi-objective optimization, these models are commonly integrated into MOEAs to replace or reduce calls to the true fitness function, forming what is known as surrogate-assisted optimization frameworks. However, ensuring high surrogate accuracy near the Pareto front remains a critical challenge, as this region is often underrepresented in training data, leading to degraded predictions precisely where they matter most (Díaz-Manríquez et al., 2016).

Therefore, the main objective of this work is to develop an ANN-based surrogate model for multi-objective optimization problems. The specific objectives are:

- i. Formulate the optimization of tryptophan production in *Escherichia coli* as a multi-objective optimization problem;
- ii. Identify the most suitable meta-models (surrogates) for this type of problem;
- iii. Implement and adapt these meta-models to the chosen problem;
- iv. Integrate the meta-models with multi-objective optimization algorithms (MOEAs);
- v. Compare the performance of the meta-models, particularly in the regions close to the Pareto front.

2 State of the art

2.1 Multi-objective Optimization and Surrogates

Multi-objective optimization (MOO) refers to problems involving multiple objective functions to be minimized or maximized simultaneously, subject to constraints, where no single solution optimizes all objectives at once (Coello et al., 2007; Deb, 2001). In contrast to single-objective optimization, MOO considers a multidimensional space where algorithms must determine the limits of the Pareto frontier and identify solutions that lie on it (Deb, 2001). A solution is Pareto-optimal when none of the objectives can be improved without degrading at least one other, and the set of all such solutions forms the Pareto-optimal set, whose projection is known as the Pareto front (Coello et al., 2007). The main goals in MOO are preserving non-dominated points, maintaining diversity along the Pareto frontier, and providing a representative set of solutions for decision making. The quality of a Pareto front approximation is assessed using performance indicators grouped into cardinality, convergence, distribution and spread (Audet et al., 2021).

Multi-Objective Evolutionary Algorithms (MOEAs) have been widely used to solve problems with multiple conflicting objectives, due to their ability to generate a set of

trade-off solutions rather than a single optimal solution (Hu et al., 2017). These algorithms explore different regions of the solution space and approximate the Pareto front, facilitating decision-making in complex problems (Zhou et al., 2011). Among them, NSGA-II, SPEA2 and MOEA/D are the most widely used (Emmerich & Deutz, 2018; Zhou et al., 2011). NSGA-II is particularly relevant due to its elitism strategy, which preserves the best individuals and improves optimization efficiency (Deb, 2001; Zhan et al., 2025).

The use of surrogate models (meta-models) has been increasingly adopted to reduce expensive fitness function evaluations (Díaz-Manríquez et al., 2016). These models approximate the objective function, reducing computational cost and the need for physical experiments. However, a critical challenge is the degradation of surrogate accuracy near the Pareto front, due to insufficient representation of extreme regions in the training data (Díaz-Manríquez et al., 2016). This limitation may reduce the effectiveness of the evolutionary search as it approaches optimal solutions. Recent studies aim to improve surrogate-assisted evolutionary algorithms (SAEAs) by incorporating local search mechanisms guided by surrogate predictions to better explore promising regions (Li et al., 2024).

Common surrogate models include Kriging, Random Forest (RF), Support Vector Regression (SVR), and Artificial Neural Networks (ANNs). These approaches differ in their modelling strategies. ANNs are computational models inspired by the functioning of the nervous system, composed of multiple layers of artificial neurons that learn highly non-linear relationships directly from data. Their flexibility and representational capacity make them particularly suited to capturing complex non-linear mappings (Wang et al., 2026), which is a desirable property in the context of surrogate-assisted optimization where the relationship between decision variables and objective values can be highly irregular. In this context, Flux Balance Analysis (FBA) is used to evaluate solutions by simulating metabolic flux distributions under stoichiometric constraints (Orth et al., 2010). However, its repeated use requires solving linear programming problems, making it computationally expensive. Therefore, FBA is often integrated with MOEAs and surrogate models to reduce computational cost (Song et al., 2025).

2.2 Tryptophan Production in *E. coli*

Tryptophan is an essential aromatic amino acid, widely used in the pharmaceutical, food and animal feed industries, with a global market exceeding 14,000 tons per year (Ren et al., 2023). Its microbial production through *Escherichia coli* fermentation has emerged as a sustainable and economically viable alternative to chemical synthesis (Niu et al., 2019). The optimization of culture conditions, including glucose supply rate, oxygen availability and genetic modifications such as gene knockouts, with the aim of maximizing tryptophan production while maintaining cell viability, constitutes a well-established problem in metabolic engineering. This framework makes it a suitable and relevant case study for the surrogate-assisted multi-objective optimization framework proposed in this work.

The optimization of tryptophan production is typically supported by the Flux Balance Analysis (FBA) method, which allows the distribution of intracellular metabolic fluxes under steady-state conditions to be predicted. This approach makes it possible to

evaluate how different environmental and genetic perturbations influence metabolic pathways and production levels.

In particular, the analysis of genes involved in the tryptophan biosynthetic pathway, as well as competing metabolic pathways, is fundamental, since genetic modifications such as gene deletions can significantly alter metabolic fluxes. These alterations may enhance tryptophan production by redirecting fluxes towards the desired pathway, or, in some cases, completely inhibit its production. Thus, the systematic exploration of these genetic and environmental factors is essential to identify optimal strategies that maximize tryptophan yield.

3 Methodology

3.1 Dataset Generation

The genome-scale metabolic model iML1515 of *Escherichia coli* (1,515 genes) was utilized to generate 4,916 valid simulations via Flux Balance Analysis (FBA). For each instance, glucose and oxygen uptake rates were randomly sampled within the range of , and between 1 and 6 simultaneous genetic knockouts were introduced.

Where is the stoichiometric matrix, is the flux vector, and are the reaction bounds. The simulation followed a two-step optimization: i) maximization of biomass; ii) maximization of tryptophan (subject to a minimum biomass constraint of 10% of the wild-type growth). Strains falling below this threshold were categorized as metabolically unviable (zero flux). Table 1 summarizes the generated dataset.

Table 1- Statistical characterization of the FBA dataset.

Dataset property	Value/Percentage
Total valid samples	4916
Metabolic viable instances	2984 (60.7%)
Unviable instances (Lethal/Zero flux)	1932 (39.3%)
Range of continuous input variables	[0.1,20.0] mmol/gDW/h
Range of simultaneous genetic Perturbations	1 to 6 random knockouts

3.2 Embedding-Based Multi-Output ANN

To map genetic knockouts and environmental conditions to metabolic fluxes, two multi-output Artificial Neural Network (ANN) surrogate architectures were developed to simultaneously estimate biomass growth rate and tryptophan production flux via a two-neuron linear output layer, differentiating strictly in their mathematical representation of genetic perturbations.

The primary approach, the Embedding-Based ANN (Figure 2a, 2b), handles genetic alterations through a dense, low-dimensional projection using a 9-feature input vector: 3 continuous variables (glucose uptake rate, oxygen uptake rate, and knockout count) and 6 categorical slots for genetic modifications. Gene identifiers were rank-based integer encoded alphabetically, mapping the non-standard gene s0001 to a fixed value of 1516, with remaining empty slots zero-padded for strains with fewer than six mutations.

To learn dense latent relationships rather than treating raw integers numerically, the 6 categorical inputs are routed through independent Embedding layers before being concatenated with the continuous variables and fed into sequential fully connected Dense layers. Conversely, the high-dimensional Multi-Hot Encoding ANN alternative (Figure 2b) expands the input space to 1,519 features, combining the 3 continuous variables with 1,516 sparse binary indicators, where 1 explicitly denotes a knockout event. This architecture completely bypasses embedding projections, routing the unified high-dimensional vector directly into a single flat input layer followed by sequential fully connected Dense layers.

To ensure a controlled comparison, hyperparameters and architectural topologies for both networks were independently tuned via Hyperopt utilizing the Tree-structured Parzen Estimator (TPE) algorithm across 200 trials. Data preprocessing, target scaling, validation constraints, and the complete optimization search space boundaries are centralized in Table 2. Final optimized configurations were evaluated against the independent test split using MAE and metrics per objective.

Table 2 - *Hyperparameter optimization search space, preprocessing scaling, and training protocols for the developed ANNs.*

Parameter	Specification/Search Space
Hidden layers	1 to 5 layers
Hidden units	16 to 512 units
Optimizers	Adam, Nadam, RMSprop
Loss function	Mean absolute error (MAE)
Validation scheme	Stratified 5-fold CV
Early stopping	Active (Patience = 25 epochs)
Activation funs	ReLu, Tanh, ELU, SELU, Softplus
Dropout rates	0.0 to 0.45
Train/Test split	85% train / 15% test
Embedding Dim.	{4,8,12,16,24,32} (<i>Embedding model only</i>)

3.3 Hybrid Classification-Regression Pipelines

To explicitly decouple phenotypic lethality from continuous flux magnitude, a two-stage hybrid framework was developed. This pipeline first passes the metabolic instances through a classification interface to gate non-viable phenotypes before routing the predicted viable instances to a downstream regression network, as illustrated in fig. 1.

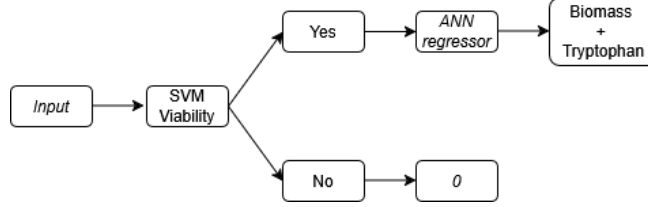


Figure 1 - Workflow of the two-stage hybrid classification-regression pipeline. The SVM filters non-viable phenotypes (0 flux assignment), routing predicted viable instances to the ANN regressor for continuous biomass and tryptophan prediction.

Classification Stage Setup

The classification interface consists of two sequential Support Vector Machine (SVM) classifiers. To ensure a controlled comparison across all hybrid variations, the internal hyperparameters of the SVMs were kept strictly static: a RadialBasis Function (RBF) kernel was utilized, regularized with a penalty parameter and a scale kernel coefficient. To mitigate the inherent dataset imbalance where viable strains represent 62.2% of the samples, a balanced class-weight correction was enforced during training. Probability estimation was enabled with a strict viability threshold set at 0.5.

Architectural Variations and Cross-Representation

Depending on the pipeline variant, the input representation was manipulated to test structural boundaries:

* **Pure Integer Hybrid:** Both stages operate on the non-positional integer-encoded arrays padded to six knockout slots.

* **Pure Multi-Hot Hybrid:** Both the SVM stage and the ANN regressor operate entirely on the 1,519 high-dimensional multi-hot encoded vector.

* **Cross-Representation (Mixed) Hybrid:** A specialized configuration where the dual-SVM interface leverages the absolute positional consistency of the multi-hot array for high-fidelity boundary definition, while the downstream ANN regressor receives the standard integer-encoded inputs.

Crucially, because the core regression task and the input dimensions for the integer-based regressor remain identical between the pure integer pipeline and the cross-representation pipeline, they share the exact same optimized hyperparameter architecture discovered during the 200-trial Hyperopt search. This isolation ensures that any observed performance leaps can be mathematically attributed solely to the classification representation acting as the independent variable.

4 Results and discussion

4.1 Dataset Generation

The generated dataset comprises 4,916 FBA simulations distributed approximately uniformly across the six knockout groups. Biomass growth rate ranges from 0 to 1.099 h^{-1} and tryptophan production flux from 0 to 8.441 mmol/gDW/h , as illustrated in Fig. 2.

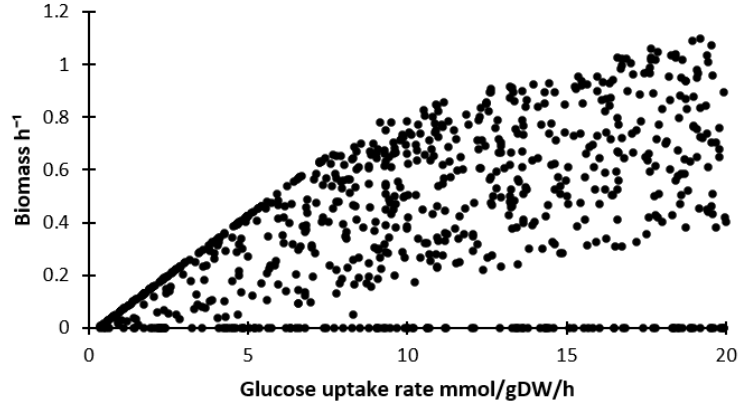


Figure 2-- Example of the distribution of biomass growth rate as a function of glucose uptake rate

Of the total simulations, 2,982 (60.7%) are metabolically viable, with viability decreasing progressively from 85.8% for single knockouts to 44.9% for six knockouts, reflecting the cumulative metabolic impact of genetic perturbation.

4.2 Results and Discussion: Surrogate Model Evaluation

The performance of the standalone neural networks and the two-stage hybrid pipelines was systematically evaluated on the independent 15% held-out test dataset. A comprehensive summary of the architectural topologies, optimization parameters, and global predictive metrics across all five configurations is centralized in Table 3.

Table 3 - Comparative performance matrix of standalone and hybrid surrogate architectures on the independent test set.

Model strategy	Input framework and topology	Viability classifier	Biomass growth rate (R^2 / Test MAE)	Tryptophan Flux (R^2 / Test MAE)
1. Standalone Embedding	9 features; 5-layers [432, 359, 287, 53, 360]	—	0.4946 / 0.10344	0.5081 / 0.76531
2. Standalone Multi-Hot	1,519 features; 5-layers [221, 319, 362, 284, 168]	—	0.8225 / 0.0710	0.8247 / 0.5410
3. Pure Integer Hybrid	Integer SVM + 3-layers [493, 17, 460]	0.6333 / 0.0063	-0.1337 / —	-0.1781 / —
4. Pure Multi-Hot Hybrid	Multi-hot SVM + 4-layers [499, 124, 382, 66]	0.9255 / 0.93	0.7180 / 0.0755	0.7071 / 0.5924

5. Cross-Rep. Hybrid	Multi-hot SVM + Embedding [493, 17, 460]	0.9300 / 0.92	0.7547 / —	0.7414 / —
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4.2.1. Standalone Architecture Performance and Representation Bias

The evaluation of standalone models revealed a profound divergence in generalization capacity driven strictly by the input encoding strategy. The Embedding-Based ANN exhibited severe overfitting and performance degradation despite its architectural complexity (a deep 5-layer topology: [432, 359, 287, 53, 360] hidden units, managed by Nadam with 24% to 31% dropout). While it achieved a remarkably low training MAE of 0.0079, it suffered a multi-fold error inflation on the test set (MAE = 0.0924), capping the at 0.4946 for Biomass and 0.5081 for Tryptophan. This collapse is tied to the high prevalence of lethal phenotypes and zero-padding, which introduced severe discontinuities that the dense embedding space failed to map linearly.

Conversely, the Multi-Hot Encoding ANN successfully bypassed these representation constraints. Utilizing a 1,519-feature input layer and an expansion-compression topology ([221, 319, 362, 284, 168] units, managed by RMSprop with a peak dropout of 37.55%), this model achieved a high and balanced above 0.82 for both metabolic targets. By ensuring absolute positional consistency, where every gene maintains a static coordinate, the network effectively resolved non-viable boundaries and learned concurrent knockout interactions without positional ambiguity, though boundary noise from true-zero flux regions still caused a minor training-to-testing MAE gap (0.0139 to 0.0645).

4.2.2 The Catastrophic Collapse of Pure Integer Hybridization

The integration of a two-stage hybrid pipeline using integer-encoded embeddings (Pure Integer Hybrid) resulted in a cascading system failure. The upstream dual-SVM classification interface displayed inadequate discriminative power, yielding a modest validation accuracy of and a test accuracy of 0.6200 for metabolic viability. Furthermore, the secondary classifier (product presence) suffered from a severe statistical artifact due to class imbalance, yielding a deceptive 0.9300 accuracy but a critical F1-score of just 0.28 for non-producing viable instances.

Because the initial classification filter failed to isolate 38% of the non-viable phenotypes, the downstream 3-layer regressor ([493, 17, 460] units, optimized via Adam was forced to allocate continuous values to lethal inputs plagued by positional ambiguity. This structural misalignment generated negative R^2 for Biomass and for Tryptophan), mathematically demonstrating that the pipeline's predictive capacity was inferior to a naive baseline mean.

4.2.3. Phenotypic Gating and Cross-Representation Synergies

The implementation of absolute positional consistency within the hybrid framework completely resolved error propagation. The Pure Multi-Hot Hybrid pipeline's primary SVM achieved outstanding discriminative power (test accuracy = 0.9255) and a

remarkably low False Positive Rate (FPR = 0.0500), successfully isolating 266 out of 280 non-viable instances. While the secondary product classifier showed a localized artifact (F1-score = 0.9856 paired with a 1.0000 FPR for non-producing viable instances due to extreme class prevalence), the primary filter successfully protected the downstream 4-layer regressor ([499, 124, 382, 66] units, with dropout peaking at 36.51% and a ReLU/SELU-to-Softplus activation sequence).

When validated strictly against the isolated, purified continuous landscape, the multi-hot regressor achieved exceptional precision. When evaluated globally across the entire independent test set using zero-flux gating for predicted non-viable instances, the integrated pipeline delivered robust metrics (MAEs of 0.0755 and 0.5924 for Biomass and Tryptophan, respectively).

Crucially, the Cross-Representation Hybrid confirmed that phenotypic classification gating is the overriding determinant of success in metabolic surrogate modeling. This architecture leveraged the 1,519-feature positional accuracy of a multi-hot SVM to isolate the lethal space (test accuracy = 0.9300, F1-scores between 0.91 and 0.94), while routing the purified continuous subset into a 3-layer embedding regressor ([493, 17, 460] units, optimized via a 6.5-hour Hyperopt search using a 4-dimensional embedding space). Backed by the high-fidelity multi-hot filter, the embedding regressor successfully generalized to unseen data, achieving a global of 0.7547 for Biomass and 0.7414 for Tryptophan. This cross-representation synergy proves that once lethal zero-flux noise is filtered, the continuous regression task becomes highly tractable even for embedding-based architectures.

4.3 Graphic visualization

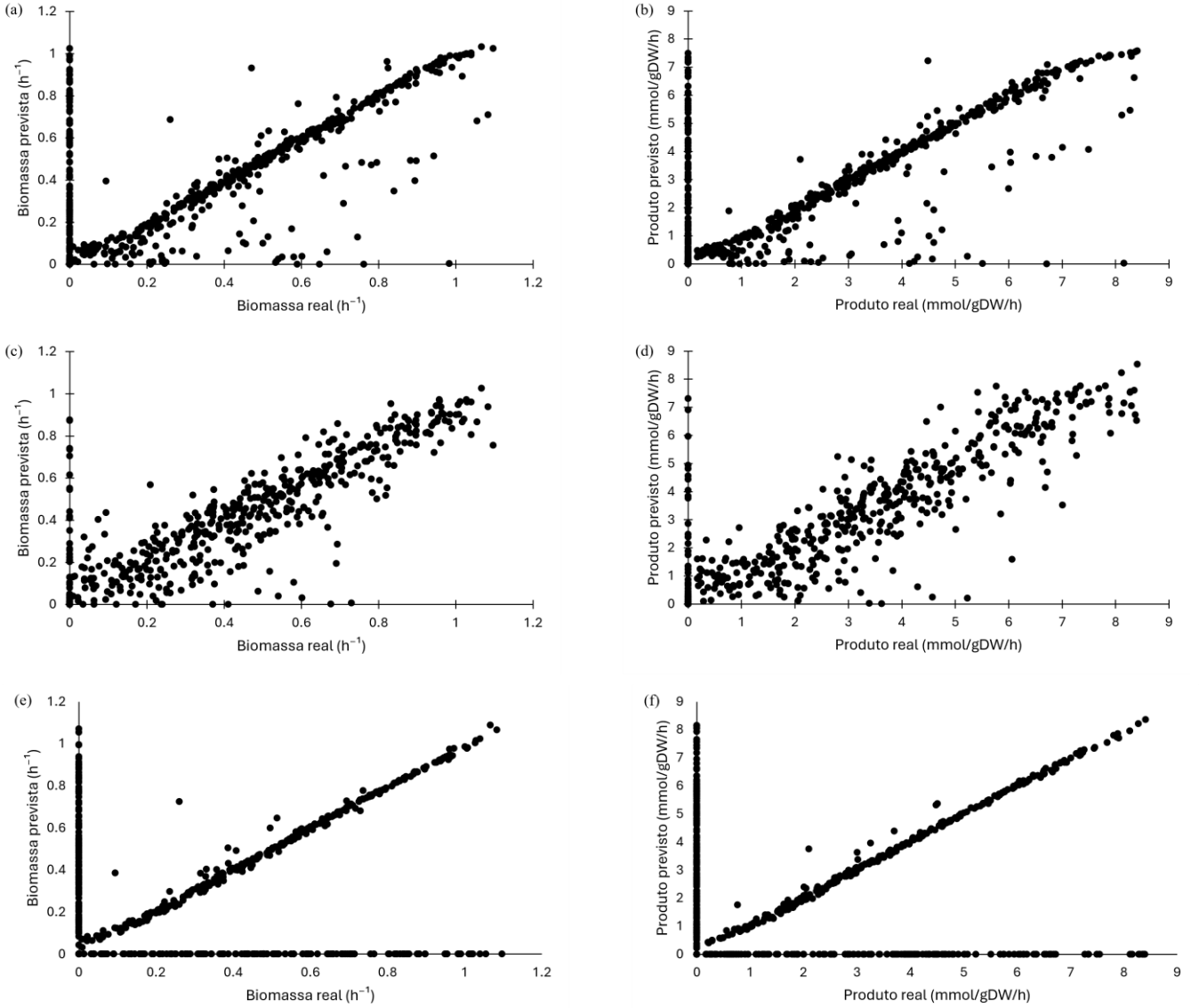
To evaluate predictive accuracy, parity plots comparing surrogate model predictions (-axis) against Flux Balance Analysis (FBA) reference values (-axis) were generated for both metabolic objectives (Figure 3). For the standalone embedding-based model (Figures 3a, 3b), viable strains exhibit exceptionally compact adherence along the identity line, demonstrating that dense latent space mapping effectively captures continuous flux magnitudes. However, the absence of a filtering mechanism forces the network to apply continuous regression logic to lethal and non-producing strains. This introduces structural limitations at phenotypic discontinuities, resulting in boundary-region error accumulation and noticeable deviations from the diagonal.

Conversely, the standalone multi-hot model (Figures 3c, 3d) exhibits complementary behavior. While binary encoding effectively captures the presence or absence of genetic perturbations, continuous flux predictions display significantly greater dispersion around the line. Because the multi-hot representation expands the input space into a high-dimensional, 1,519-feature sparse matrix, the dense neural network faces increased geometric complexity when stabilizing its regression, resulting in lower precision and a less compact distribution than the embedding-based counterpart.

Figures 3e–3h illustrate the integration of an upstream Support Vector Machine (SVM) classifier for phenotypic gating, which isolates lethal and non-producing instances to anchor their predictions at the zero coordinate. Consequently, the pure integer hybrid (Figures 3e, 3f) achieves substantially tighter adherence to the identity line. In contrast, while the pure multi-hot hybrid (Figures 3g, 3h) excels at identifying unviable

phenotypes, its downstream regressor exhibits a systematic skew, overpredicting low values near the origin and underpredicting higher fluxes, which increases diagonal deviation.

Finally, the cross-representation hybrid (Figures 3i, 3j) represents the methodological culmination of this work, combining the multi-hot array's absolute positional consistency for high-fidelity SVM survival gating with the embedding network's superior continuous flux regression. This configuration yields the cleanest, most precise prediction distribution: biomass predictions (Figure 3i) display an ultra-compact, linear alignment along the diagonal, while tryptophan predictions (Figure 3j) cleanly resolve the zero-inflation problem with an explicit cluster at and accurate alignment for producing strains. Visually, these results confirm that filtering non-viable noise allows the embedding regressor to operate exclusively within its optimal continuous domain, unlocking the maximum generalization potential of the hybrid framework.



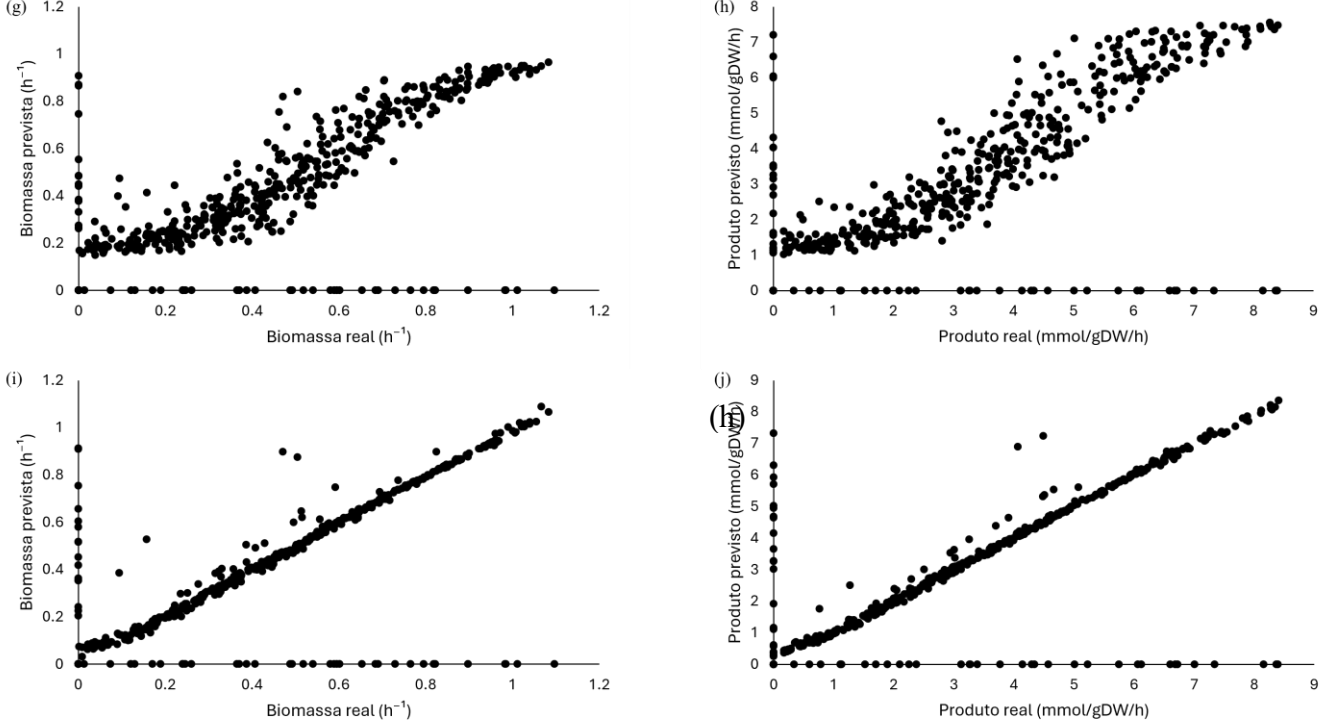


Figure 3 - Parity plots comparing surrogate model predictions (-axis) against FBA reference values (-axis) for biomass growth (left panels: a, c, e, g, i) and tryptophan production (right panels: b, d, f, h, j). Evaluated architectures comprise: (a, b) standalone embedding-based ANN; (c, d) standalone multi-hot ANN; (e, f) pure integer hybrid pipeline; (g, h) pure multi-hot hybrid pipeline; and (i, j) cross-representation hybrid pipeline. The dashed line denotes the identity line.

5 Conclusions

This work successfully validated the development of Artificial Neural Network (ANN)-based surrogate models as a computationally efficient alternative for accelerating the multi-objective optimization of tryptophan production in *Escherichia coli*. Using the genome-scale metabolic model iML1515, the main challenge was to overcome the highly irregular and discontinuous topology of the solution space, which was severely affected by the presence of lethal (non-viable) phenotypes.

The comparative analysis showed that the two input representations exhibit complementary strengths and weaknesses. Integer encoding with embeddings provided superior alignment of flux predictions with the identity line but struggled with the discontinuities associated with lethal phenotypes. In contrast, multi-hot encoding mapped viability boundaries more effectively, although its high-dimensional sparse representation increased dispersion in continuous regression tasks.

The proposed mixed hybrid architecture successfully resolved these limitations by combining a multi-hot SVM for phenotypic gating with an embedding-based ANN for flux prediction. By filtering non-viable strains before regression, the pipeline eliminated a major source of noise and achieved the most accurate predictive distribution observed in this study.

By successfully mitigating the zero-inflation challenge through this cross-representation approach, this work completes its foundational development phase. The resulting hybrid architecture provides a highly precise, robust, and computationally efficient surrogate engine that is now fully validated and prepared to be integrated into multi-objective evolutionary optimization frameworks (MOEAs) as the immediate next step of this research line.

6 Future work

Although the proposed hybrid surrogate modelling framework demonstrated high predictive accuracy and robustness, several opportunities remain for further improvement. First, future work should focus on optimizing the Support Vector Machine (SVM) classifiers used for phenotypic gating. A systematic hyperparameter optimization strategy, including the exploration of different kernel functions and the tuning of parameters such as the regularization coefficient (C) and kernel-specific parameters (e.g., γ), could further improve the discrimination between viable and non-viable strains, as well as between producing and non-producing phenotypes. Optimization techniques such as grid search, Bayesian optimization, or evolutionary algorithms could be employed to identify the optimal classifier configuration.

In addition, alternative classification algorithms, including ensemble methods such as Random Forests, Gradient Boosting, or deep neural network classifiers, could be investigated as potential replacements for the SVM gating stage. Comparing these approaches may provide insights into their relative ability to capture complex phenotypic boundaries.

Future studies could also explore the optimization of the ANN architectures used for flux prediction. Hyperparameters such as the number of embedding dimensions, network depth, number of neurons per layer, activation functions, regularization techniques, and learning rate schedules could be systematically evaluated to further enhance predictive performance and generalization capabilities.

Another important direction would be to assess the transferability of the proposed framework to other genome-scale metabolic models, target metabolites, and microbial hosts. Validating the methodology across different metabolic engineering scenarios would provide stronger evidence of its general applicability and robustness.

Building directly upon the validated framework developed in this work, the immediate next phase of this research consists of integrating the surrogate models directly within multi-objective optimization frameworks, such as evolutionary algorithms (e.g., NSGA-II). This will enable the efficient exploration of larger design spaces while significantly reducing the computational burden associated with repeated Flux Balance Analysis (FBA) simulations, effectively achieving the goal of this optimization pipeline.

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